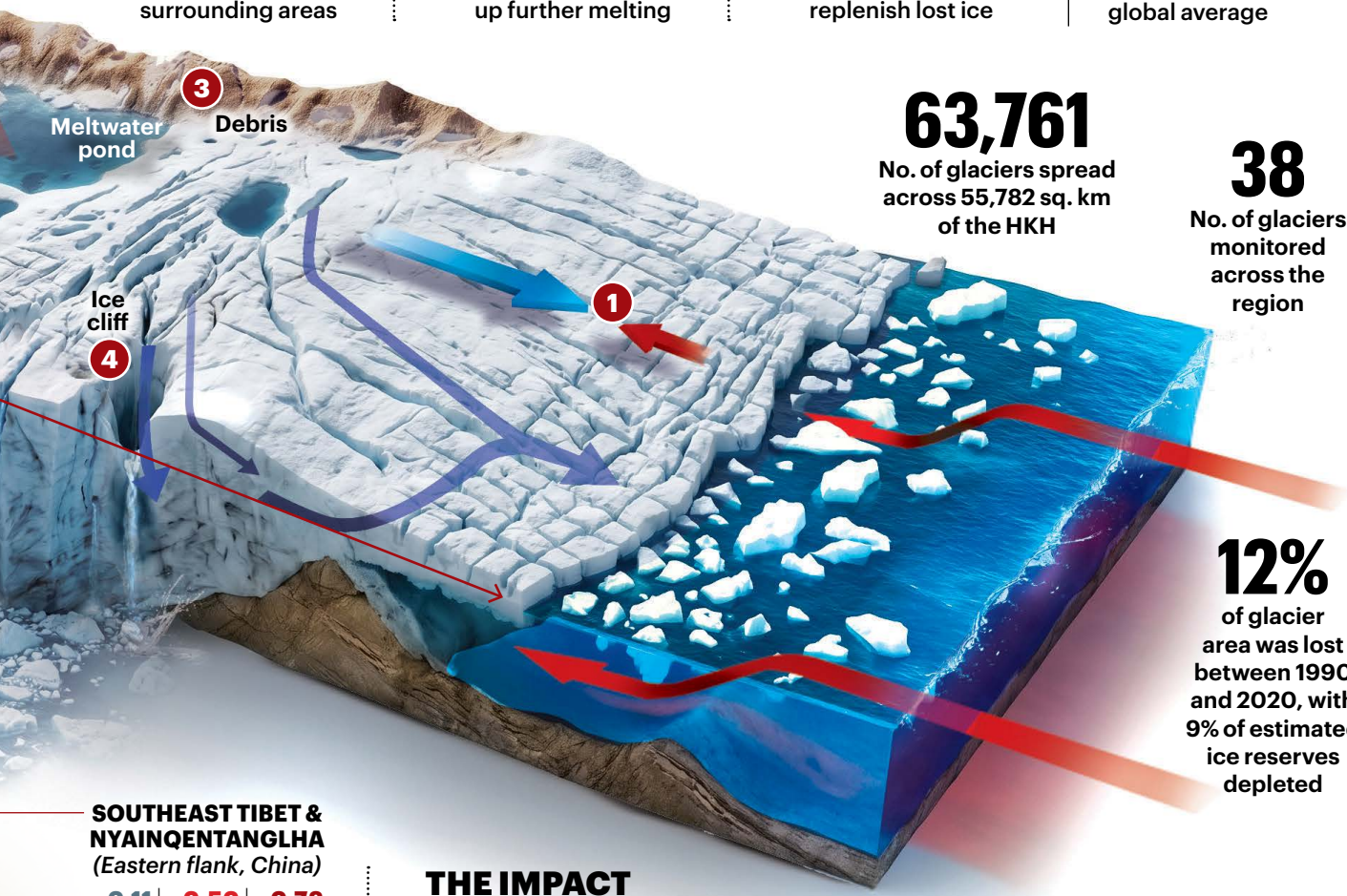
HOW THE GLACIERS MELT

1. Rising temperatures melt exposed ice first at the glacier's lower tongue, causing visible retreat over time
2. Faster warming at higher altitudes reduces snowfall while accelerating melt, squeezing glaciers from both ends
3. Rocky debris can either speed up melting when thin or slow it down when thick, creating mixed effects
4. Exposed ice cliffs and meltwater ponds form on glaciers, melting ice much faster than surrounding areas
5. As bright ice melts, darker surfaces absorb more heat, creating a cycle that speeds up further melting
6. Rising temperatures push the snowline higher, reducing the glacier's ability to replenish lost ice

WHY THEY ARE MELTING

- The HKH has warmed at 0.2°C per decade over the past 50 years
- Even at 1.5°C global warming, the region is projected to heat up by 1.8 ± 0.4°C
- High mountain zones are warming faster than the global average



63,761
No. of glaciers spread across 55,782 sq. km of the HKH

38
No. of glaciers monitored across the region

12%
of glacier area was lost between 1990 and 2020, with 9% of estimated ice reserves depleted

SOUTHEAST TIBET & NYAINQENTANGLHA
(Eastern flank, China)
−0.11 | −0.50 | −0.78
The worst-performing region across all periods: Warm, wet, monsoon-fed maritime glaciers with high sensitivity to temperature

WEST NYAINQENTANGLHA
(Central-east Tibet)
−0.19 | −0.53 | −0.66
Severe and accelerating

THE IMPACT

The Himalayan system feeds three major river basins in India, with millions of people dependent on them. How they will be affected

INDUS BASIN

Covers: *Ladakh, J&K, Himachal Pradesh, Punjab*

- Erratic water supply for high-altitude areas
- Higher risk of flash floods and landslides
- Hydropower instability

GANGA BASIN

Covers: *Uttarakhand, Uttar Pradesh, Bihar, Delhi*

- Increased glacial lake outburst floods
- More disasters like that hit Kedarnath
- Seasonal water instability

BRAHMAPUTRA BASIN

Covers: *Northeast region*

- Glacier-fed streams becoming unstable
- Flood intensity worsens
- Rivers may lose perennial character in some stretches

Source: 'Changing Dynamics of Glaciers in the Hindu Kush Himalaya Region from 1990 to 2020' and 'HKH Glacier Outlook 2026' by the International Centre for Integrated Mountain Development